

Title	CLOUD TECHNOLOGIES
Subject Code	23BCA4I02
Hours / Week	3
Credits	3

COURSE OBJECTIVES	
COB1	Classify cloud computing technologies and its models along with services, types and challenges with cloud applications.
COB2	Explain the basic concepts and terminology of virtualization and cloud technologies in current environment.
COB3	Examine and describe the functions of virtualization and deploy the concepts of virtualization technologies along with managing the virtual machines.

COURSE OUTCOMES	
CO1	Discuss about the Cloud computing technologies and its models.
CO2	Explain the basic concepts and terminology of virtualization and cloud technologies in the current IT environment.
CO3	Examine the functions of Virtualization in cloud computing environment.
CO4	Illustrate the Azure Basics, Services, Portals, Management and Virtual Machine Management.
CO5	Describe the basic concepts of Security essentials in Cloud and Virtualization.

SYLLABUS				
Module (Hours)	Contents	Tools Used / Assessment and Activity	CO Mapping	PO Mapping
Module 1 (09 Hrs)	Introduction Cloud Computing Introduction to Cloud Computing, History and Evolution of Cloud Computing, Types of clouds, Private Public and hybrid clouds, Cloud Computing Architecture, Cloud computing infrastructure, Merits of Cloud computing, Cloud computing delivery models and services (IaaS, PaaS, SaaS), Infrastructure as Cod (IaC), obstacles for cloud technology, Cloud vulnerabilities, Cloud challenges, Practical applications of cloud computing.	Internal Continuous Assessment – Periodic Internal Test1, Preparatory Exam Activity 1 – Linkedin Course	CO1	P01 P02 P03 P04 P05 P06 P07
Module 2 (09 Hrs)	Introduction to Virtualization Define Virtualization, Need Virtualization, Virtualization in Current IT scenario, Comparing Virtualized and Non- Virtualized Environment. Understanding Virtualization Technologies: Server Virtualization - Operating system Virtualization, Hardware Emulation and Para-virtualization, Client Virtualization - Application Packaging, Application Streaming and Hardware emulation and Storage Virtualization.	Internal Continuous Assessment – Periodic Internal Test1, Preparatory Exam Activity 1 – Linkedin Course	CO2	P01 P02 P04 P06 P07 P09 P010
Module 3 (09 Hrs)	Virtualization Technology terms & techniques Benefits of VDI, Introduction to virtual machine, various components of virtual machine to be considered while creating a virtual machine, what is virtual hard disk and its usage in a different virtualized environment. Different types of Hypervisor management application.	Internal Continuous Assessment –Preparatory Exam. Activity 1 – Linkedin Course Activity 2 – case study	CO3	P01 P02 P04 P05 P06 P07 P011
Module 4 (09 Hrs)	Virtualization and the Cloud Visualizing Virtualization: Characteristics, using a hypervisor in virtualization, abstracting hardware assets, Managing Virtualization: foundational issues, abstraction layer, provisioning software,	Internal Continuous Assessment –Preparatory Exam. Activity 1 –	CO4	P01, P02 P03, P04 P05, P06 P07, P010

	virtualizing storage, hardware provisioning, Taking virtualization into the cloud Considerations	Linkedin Course Activity 2 – case study		P011, P012
Module 5 (09 Hrs)	Security Essential in Cloud and Virtualization Vulnerabilities and Attacks, Hypervisor Security, Storage, Hardware, and Network, Cloud Security Mechanisms, Data Security, Digital Signature, Hashing, Virtualization Security, Privacy and Security in Cloud Storage Services, Cloud Data Protection Models, Security and Privacy Properties in Multiclouds.	Internal Continuous Assessment –Preparatory Exam. Activity 1 – Linkedin Course Activity 2 – case study	CO5	P01, P02 P03, P04 P05, P06 P07, P08 P09, P010 P011, P012

Text Book:	
1. Ivanka Menken, Gerard Blokdijk, —Virtualization--the complete cornerstone guide to virtualization best practices Lightning Incorporated, 2008.	Source
2. Chris Wolf, Erick M. Halter, —Virtualization: From the Desktop to the Enterprise , EBook, 2005.	
3. Ray J Rafaels, —Cloud Computing: From Beginning to End , 1st April, 2015.	
4. Judith Hurwitz, Robin Bloor, Marcia Kaufman & Fern Halper, Cloud Computing for Dummies, Wiley Publishing, 2010.	
Reference books:	
1. Matthew Portnoy, Virtualization Essentials, edition-2 (7 th Oct 2016).	
2. Cloud Computing: Principles and Paradigms, Rajkumar Buyya, James Broberg, Andrzej M. Goscinski,, John Wiley and Sons Publications, 2011.	
3. Architecting the Cloud: Design Decisions for Cloud Computing Service Models (SaaS, PaaS, and IaaS) by Michael Kavis, 28 January 2014.	

Web Links	
1. Benefits of virtualization-	https://searchdatacenter.techtarget.com/feature/Case-studies-show-the-benefits-of-virtualization
2. learn Virtual Machines RIGHT NOW -	https://www.youtube.com/watch?v=wX75Z-4MEoM1
3. What is a Hypervisor? -	https://www.youtube.com/watch?v=LMAEbB2a50M
4. Cloud Computing Tutorial for Beginners	https://www.youtube.com/watch?v=usYySG1nbfl